Simulating Mechanical Energy Transfer in Rowing

A Cycle-Resolved Model and a Framework for Disturbance Analysis

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Abstract

Rowing propulsion varies throughout the stroke, while hull resistance and crew motion affect how the applied force translates into boat speed. I examine this interaction using a cycle-resolved surge model and a proposed moving-mass and pitch extension. The baseline represents the boat and crew as a single mass under prescribed periodic thrust and quadratic drag. In an illustrative 20-second startup, the computed average speed during the latter half is 5.785 m/s, and the kinetic-energy gain is 6.49% of modeled thrust work. This ratio characterizes the chosen startup interval and is not rowing efficiency. Timestep refinement reduces the energy-balance residual, and a separate moving-mass implementation passes momentum and numerical consistency checks. Synthetic sensor and parameter-fitting examples illustrate a possible measurement workflow and its sensitivity to model assumptions. These results are simulations, not observations of a crew. The framework identifies measurements and validation steps needed before drawing conclusions about rowing technique or performance.

1 Background and scope

Rowing involves moving body segments, oar-water interactions and hull motion that cannot be resolved by this single-mass model. A biomechanical review by Baudouin and Hawkins discusses performance factors and velocity fluctuations [1]; it provides context rather than validation of the present parameter choices. Hill and Fahrig examined how within-stroke velocity fluctuations can affect race time using a mathematical approximation [3]. Mola develops a higher-fidelity treatment of rowing-boat dynamics [4], illustrating the additional modeling detail beyond the present surrogate. A useful next step is to fit time-resolved thrust and speed to independent measurements, assess parameter uncertainty, and compare predictions with held-out strokes. A longer simulation with a cycle-to-cycle convergence criterion would distinguish startup behavior from periodic operation. Comparisons of stroke rates should specify whether mean thrust, work per stroke, or average power is held fixed.

2 Model

The boat and crew are represented by a single mass m with speed v. The equation of motion is

$$m\frac{dv}{dt} = F(t) - kv^2. \tag{1}$$

For the baseline, $m = 766.5$ kg, $k = 40$ kg/m, and the prescribed thrust is 3000 N for 0.9 seconds in each 2-second cycle, followed by zero thrust. The boat starts from rest. These values are illustrative model inputs, not measurements from a crew.

The model calculates thrust work $W = \int Fv\,dt$, drag work $D = \int kv^3\,dt$ and the change in kinetic energy $\Delta K = m(v_f^2 - v_0^2)/2$. In the continuous model $W - D = \Delta K$. Thrust work on the modeled mass is not the rowers' total mechanical or metabolic work. The ratio $\Delta K/W$ describes how much input work remains as kinetic energy at the end of this startup interval; it is not rowing efficiency and depends on the interval chosen.

2.1 Stroke cycle and energy accounting

With cycle period $T = 2$ s and drive duration $d = 0.9$ s, the prescribed force is F_d when $0 \leq t \bmod T < d$, and zero otherwise. The resulting duty fraction is $d/T = 0.45$. The force acts on the combined modeled mass; it is not a direct measurement of force at an oar handle or a physiological power estimate.

Multiplying the equation of motion by v and integrating gives the continuous energy balance $W - D = \Delta K$. The discrete calculation evaluates force at the new grid time and power using the previous velocity; the squared velocity update introduces a step-dependent residual. A small residual is a necessary consistency check, but does not verify the drag coefficient or thrust model. Velocity is clipped at zero in the code, so an energy balance would also need to account for that operation if it became active. It is not active in the reported baseline.

2.2 Numerical formulation and sensitivity design

For the baseline, the sampled force is evaluated at t_i and applied over the preceding velocity update. With v_{i-1} denoting the previous speed, the implemented step is

$$v_i = \max\left[0, v_{i-1} + \frac{\Delta t}{m}(F(t_i) - kv_{i-1}^2)\right]. \tag{2}$$

The cumulative thrust and drag work use the same force samples and previous speeds. Without clipping, substituting this update into the kinetic-energy difference gives an additional term $\frac{1}{2}m(v_i - v_{i-1})^2$ at every step. Consequently, finite-step work and kinetic-energy differences need not balance exactly. The reported residual and timestep comparison quantify this discretization error for the chosen baseline. Stroke-transition timing can also affect results; a convergence check should refine both the timestep and the representation of the drive and recovery intervals.

The sensitivity scenarios vary one input in the supplied script at a time: drag from 40 to 20 kg/m, mass from 766.5 to 383.25 kg, and period from 2.0 to 1.5 s. In the period case, the 0.9 s drive is held fixed, so the duty fraction increases from 0.45 to 0.60. The mean applied force over a complete cycle therefore also increases. These comparisons isolate code inputs but not necessarily a physiological or equal-work intervention. The last-half average is a finite-window statistic, and the endpoint sample is included in the discrete mean. A cycle-convergence criterion would be needed to characterize a periodic regime independently of the chosen 20-second horizon.

3 Method and results

The equation is integrated with explicit Euler steps of 0.01 seconds for 20 seconds. The baseline average speed over the latter half is 5.785 m/s and the maximum is 6.759 m/s. Thrust work is approximately 141.2 kJ, drag work 132.3 kJ and kinetic energy gain 9.17 kJ. The gain is 6.49%

of thrust work. The difference of about -316 J in $W - D - \Delta K$ is a numerical residual from the discrete scheme.

At a 0.005-second step the residual is about -158 J; at 0.001 seconds it is about -31 J. The corresponding latter-half average speeds are 5.782 and 5.781 m/s. This convergence check addresses discretization, not the physical accuracy of the inputs.

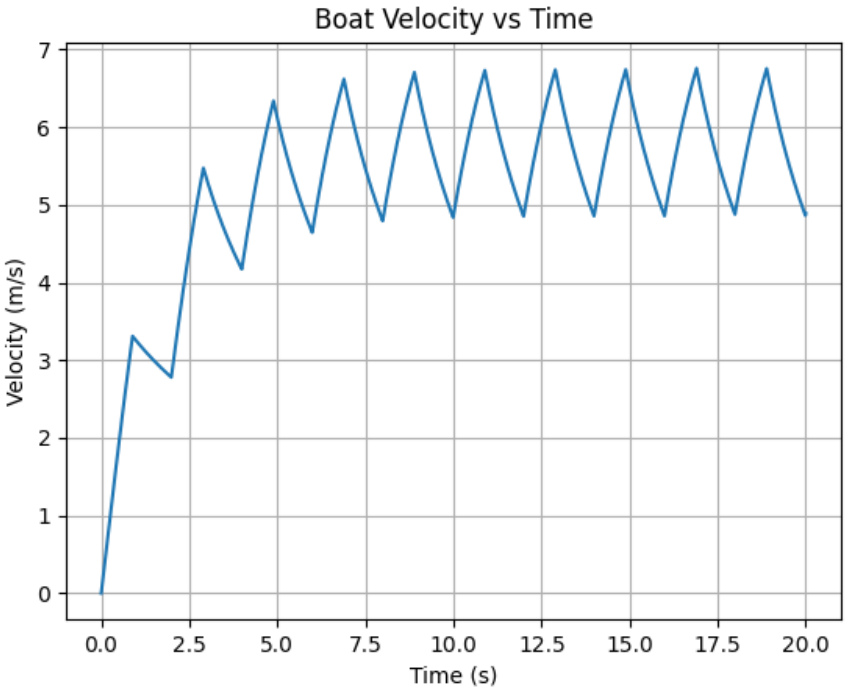


Figure 1: Baseline speed during the 20-second simulation.

3.1 Parameter comparisons and limitations

With the default step, halving drag raises the latter-half average speed to 8.162 m/s; halving mass gives 5.719 m/s over the same finite interval. Shortening the cycle from 2.0 to 1.5 seconds gives 6.673 m/s, but keeping the drive at 0.9 seconds also increases the drive fraction and input. This is therefore not an isolated comparison at constant work or power.

The model excludes blade slip, rower motion relative to the shell, varying force profiles, wind and waves. It has not been calibrated or compared with measured trajectories. More realistic thrust and drag measurements, timestep checks and experimental validation would be needed before applying the numerical predictions to a crew.

3.2 Sensitivity profiles

Figures 2–4 show the complete transient profiles rather than only the summary averages. The mass comparison changes acceleration and stroke-to-stroke fluctuations as well as the finite-interval average. The 20-second averages do not prove that a periodic steady state has been reached. Lower drag changes the relationship between thrust, speed and work, so it is not a comparison at equal

mechanical power. The higher-rate case uses a longer fraction of each cycle for the drive and therefore changes more than stroke frequency.

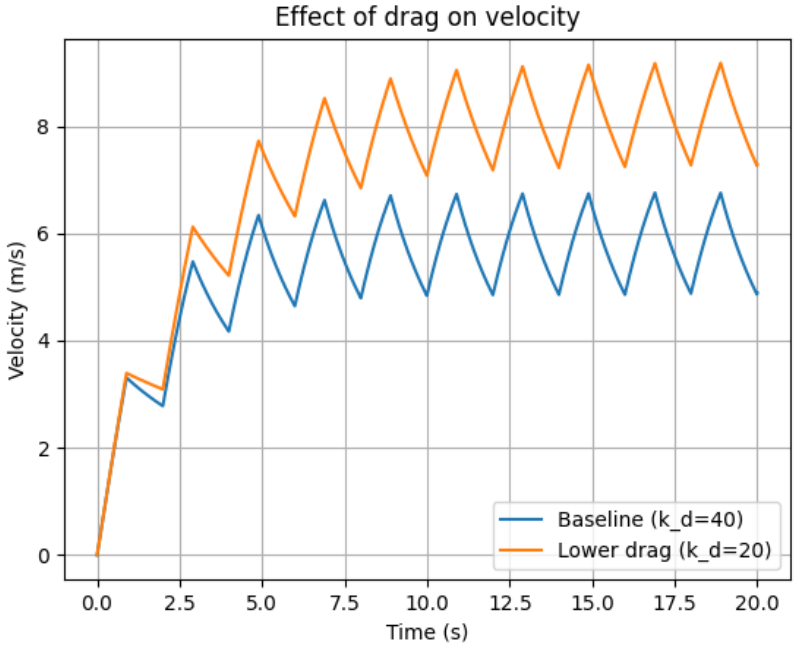


Figure 2: Velocity for baseline and lower drag.

4 Moving-mass extension

The lumped model makes the hull and crew move as one mass. To examine recovery motion, I consider a reduced model in which the crew center of mass moves relative to the hull. Let m_b be the hull-fixed mass, m_r the moving mass, $M = m_b + m_r$, X_b the hull position, and x_r the crew position relative to the hull. The combined center of mass is

$$X_c = X_b + \frac{m_r}{M} x_r. \tag{3}$$

If external surge forces are represented by thrust F and drag D, differentiating twice gives

$$M\ddot{X}_b = F - D - m_r \ddot{x}_r. \tag{4}$$

A prescribed internal acceleration changes hull acceleration even though it does not change total momentum in an isolated system. The sign depends on the coordinate direction. The relation follows from the two-mass assumptions; it does not determine crew trajectory, blade force, or internal work. Measured or explicitly prescribed motion would be needed to evaluate the term. The appendix's prescribed trajectory uses a half-cosine drive followed by a cosine, fifth-order polynomial, or blended recovery. I reconstructed this component as `src/slide_kinematics.py`; tests compare its analytic derivatives with finite differences away from transitions. Position and velocity are continuous at the drive–recovery boundary, but acceleration may jump there. This component alone does not reproduce the original multibody or IMU figures.

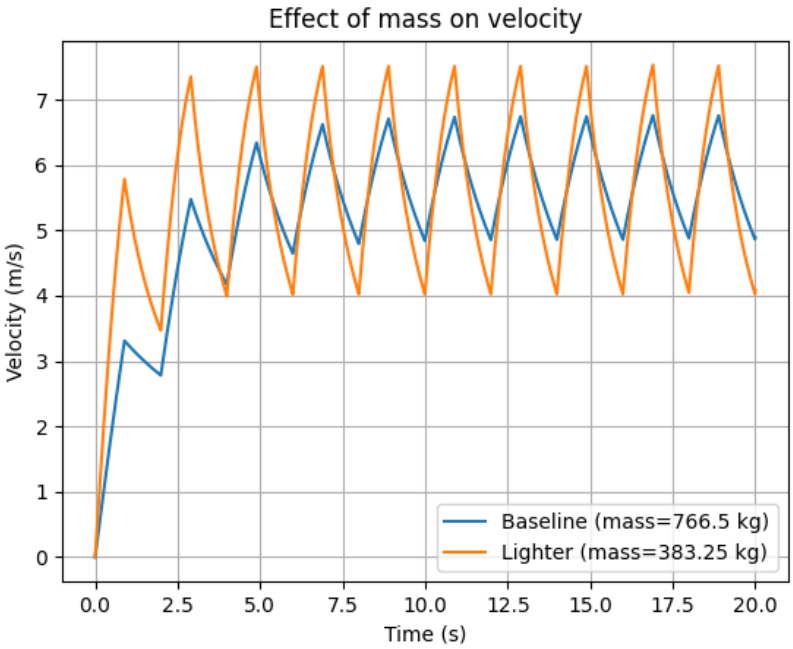


Figure 3: Velocity for baseline and lower mass.

4.1 Pitch and disturbance resistance: a proposed surrogate

A possible extension represents pitch θ with effective inertia, damping, and restoring stiffness:

$$I_\theta \ddot{\theta} + c_\theta \dot{\theta} + k_\theta \theta = \tau(t). \tag{5}$$

A phenomenological drag law can then be written $D = kv^2(1 + \alpha\theta^2 + \beta\dot{\theta}^2)$ for forward motion. The coefficients must carry units consistent with the dimensionless multiplier and must be fitted or independently supported. The incremental work $\int kv^3(\alpha\theta^2 + \beta\dot{\theta}^2)dt$ compares terms at the same instantaneous speed. It is not automatically the difference between two coupled simulations, because changing drag can also change speed and pitch. These equations are implemented in a separate reconstructed research prototype using the parameter values in the appendix. The reconstruction passes unforced momentum and numerical refinement checks. It has not reproduced the original synthetic figures or been experimentally validated; its prescribed kinematics and pitch law are hypotheses.

4.2 Speed fluctuations and power

For $v = \bar{v} + u$, with $\langle u \rangle = 0$, the exact identity is

$$\langle v^3 \rangle = \bar{v}^3 + 3\bar{v}\langle u^2 \rangle + \langle u^3 \rangle. \tag{6}$$

For nonnegative speeds, convexity implies $\langle v^3 \rangle \geq \bar{v}^3$. Under this model a variable speed at fixed mean speed and fixed drag coefficient requires at least as much average drag power as constant speed. This comparison does not establish that reducing hull speed fluctuations always improves race performance: changing technique may alter propulsion, crew motion, and other losses. The model makes the energetic assumption explicit rather than providing a coaching prescription.

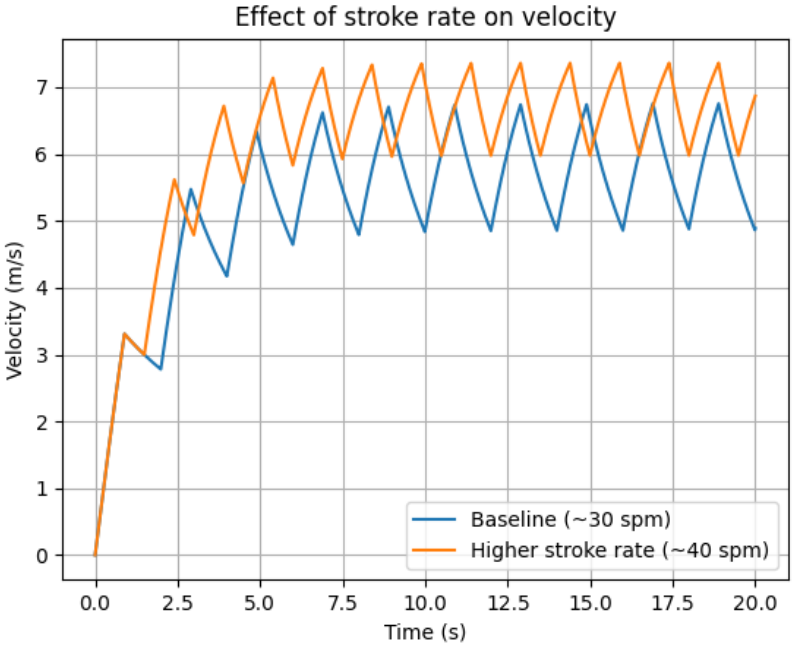


Figure 4: Velocity for two cycle periods with fixed drive duration.

4.3 Extended model and uncertainty

The moving-mass formulation separates the speed of the hull from the speed of the total center of mass. This distinction matters when interpreting a hull-mounted sensor: the internal acceleration term can cause a hull-speed fluctuation without creating or destroying total momentum. The model still prescribes the crew motion rather than solving for muscle, seat, foot-stretcher and oar forces. Its pitch response is an effective oscillator, and the pitch-dependent drag expression is a parameterization rather than a measured hydrodynamic law.

The archived manuscript contains more than one simulator. One version uses a moving-mass term in surge and a pitch moment tied to slide position; another uses a different motion and a pitch moment tied to crew acceleration, without the same surge coupling. Their parameter sets and force profiles differ, so results from the two formulations should not be pooled. The present reconstructed implementation follows the first family with adaptive time integration. The alternative archived beta-pulse routine normalizes by the maximum of the supplied array; in the listed scalar ODE call that can produce a constant normalized force or an undefined value at zero. The model comparisons here therefore do not use that routine.

The recovered IMU methods also represent different levels of approximation. The implemented filter integrates small-angle roll and pitch rates and corrects them toward an accelerometer-derived gravity direction. Its interpretation depends on sensor placement and the assumption that acceleration supplies a useful gravity reference. Dynamic acceleration can contaminate orientation estimates. The original source describes a separate quaternion-based pipeline, but that is not equivalent to the small-angle routine and has not been reconstructed here. The synthetic sensor and filter in this project share a positive vertical gravity-at-rest convention, addressing the sign mismatch between the earlier synthetic generator and its filter; this is a convention choice, not validation of the sensor model.

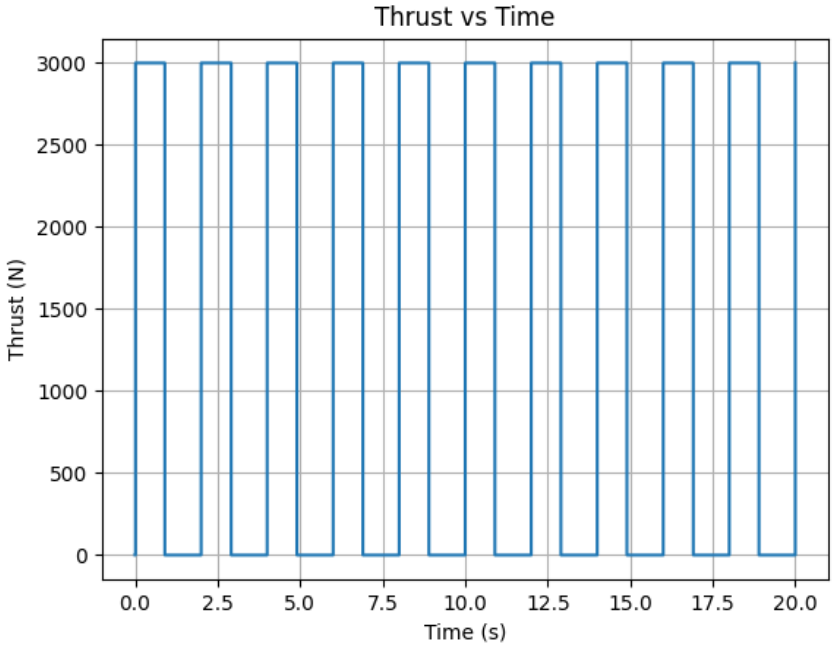


Figure 5: Prescribed thrust profile.

5 Measurement and system identification

To connect the model to measurements, a future protocol could synchronize a hull-mounted inertial measurement unit, independent speed measurements, stroke events, and a force or power proxy. Sensor orientation, gravity removal, mounting, calibration, sampling and filtering should be documented before calculating acceleration or pitch features. Integrating acceleration to obtain speed requires independent drift correction. A rowing sensor review discusses measurement choices and their constraints [2]. Brice and colleagues assessed inertial measurements of torso and pelvis angles in elite rowers against a reference system [5]; this supports further measurement work but does not validate a hull disturbance index. Events detected from a hull signal should be checked against an independent stroke reference rather than assumed to coincide with the catch or finish.

Let y_i denote measured waveforms or features and $\hat{y}_i(\psi)$ their model counterparts. A candidate fit minimizes $\sum_i (y_i - \hat{y}_i)^T \Sigma_i^{-1} (y_i - \hat{y}_i) + \lambda \Omega(\psi)$ over physically admissible parameters. The covariance or weighting matrix Σ_i and regularizer Ω must be specified before fitting. Sensitivity rank, profile likelihood or comparable diagnostics are needed to examine whether parameters can be separated; a good waveform fit alone does not prove identifiability. Validation should use held-out sessions or trials, report uncertainty and compare predictions with independently measured outcomes.

5.1 Synthetic reconstruction and checks

The source manuscript's synthetic demonstration is distinct from such empirical validation. I reconstructed uniform resampling and a small-angle complementary filter from its listing, with tests for stationary tilt and interpolation. The original listing's synthetic accelerometer uses a negative gravity sign while its preprocessing equations assume positive vertical gravity at rest. The reconstructed synthetic sensor adopts the filter's positive vertical gravity at rest; a stationary test

checks this convention. Its small-angle planar acceleration approximation is not a validated full inertial sensor model.

A reconstructed acceleration-magnitude peak detector was checked on regularly spaced synthetic pulses; its peaks are candidate events, not validated catch labels. A separate least-squares prototype fits pitch-drag and trim coefficients to modeled speed and pitch-rate observations. For a four-second noiseless simulation generated by the same model, it recovered the inputs 2.5 and 18000 to the printed precision. This inverse-crime test checks implementation only; it does not establish parameter identifiability under noise, model mismatch, or real measurement conditions. In one fixed-seed synthetic trial with speed noise standard deviation 0.01 m/s and pitch-rate noise 0.002 rad/s, the fitted pitch-drag coefficient was 2.31 for a true value of 2.5. With the same data and the drag coefficient fixed 10 percent too high, the fitted coefficient fell near zero, while the optimizer still reported success. This single example illustrates model dependence, not a statistical uncertainty interval.

The synthetic trajectory in Figure 6 has large pitch rates and even brief backward hull motion from rest; it is illustrative, not a realistic prediction. The source manuscript's original synthetic figures have not been independently reproduced. An adapted recovery-blend sweep follows the listing's three plotted metrics rather than the cost-per-distance quantity described in the original caption. In this reconstruction, increasing the blend from 0 to 1 increases the computed jerk RMS, pitch RMS, and speed coefficient of variation. These outcomes depend on the prescribed trajectory, model parameters, and numerical timestep; the acceleration discontinuities also make numerical jerk sensitive to sampling. They do not establish a technique recommendation.

The original cohort example ranks recordings by the mean of an index standardized separately within each recording. That mean is zero to numerical precision, so the resulting ordering does not establish differences between athletes. A separate archived beta-force implementation normalizes each input array by its own maximum; called on one time point at a time, it becomes constant at nonzero phases and undefined at zero. I therefore do not use that version for the simulations presented here.

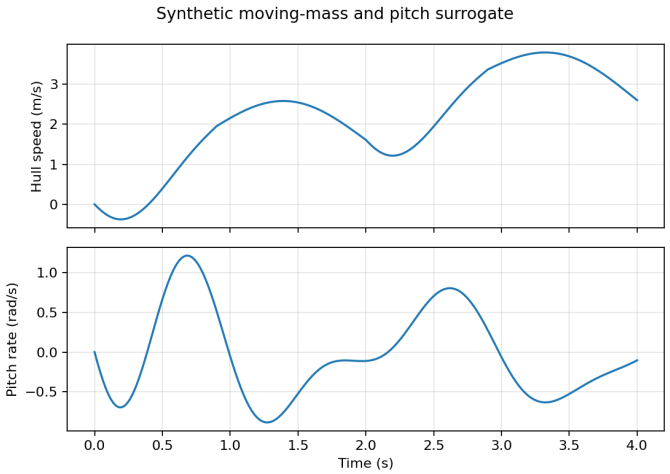


Figure 6: Synthetic trajectory from the reconstructed moving-mass and pitch surrogate, using illustrative manuscript parameters.

5.2 Synthetic sensitivity and implementation checks

The recovery-blend calculation varies the relative contribution of cosine and fifth-order polynomial recovery trajectories. The recovered plotting script evaluates hull-acceleration jerk RMS, pitch RMS, and the coefficient of variation of speed. In the adapted example the respective values change from approximately 10.04 to 16.05 m/s^3, 0.1413 to 0.1452 rad, and 0.1853 to 0.1915 as the blend rises from zero to one. The acceleration can be discontinuous at a stroke transition, so a numerically differentiated jerk is especially sensitive to timestep and should not be interpreted as a measured athlete characteristic. Figure 7 separates the three quantities instead of plotting their unlike units on one axis.

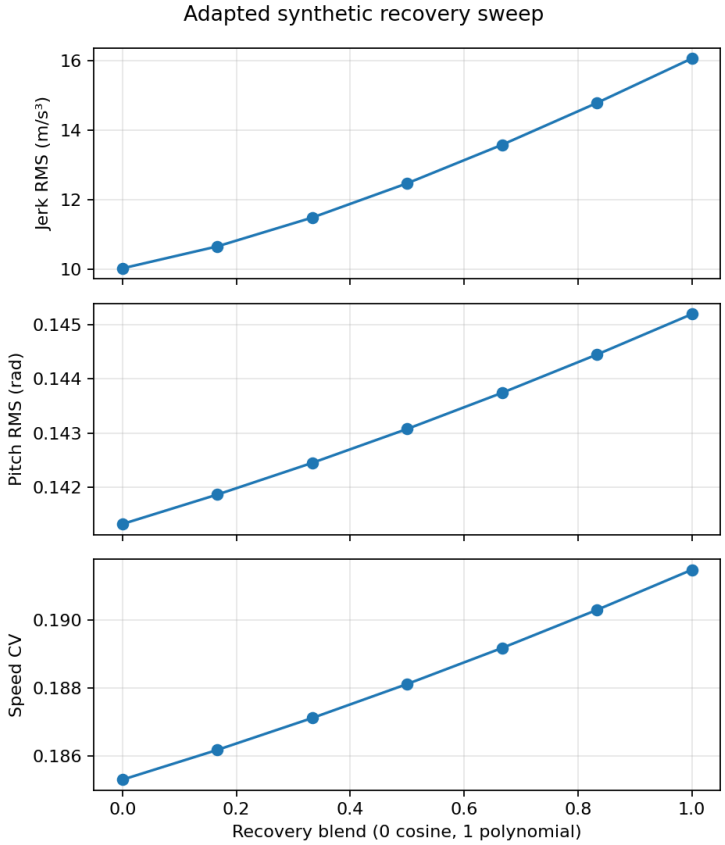


Figure 7: Adapted synthetic recovery-blend sweep. The plotted metrics are not disturbance work per distance.

The synthetic sensor demonstration adds prescribed bias, random-walk drift and noise to simulated channels. Both traces in Figure 8 are model-generated; the perturbed trace is not a recording from a real crew. Candidate event detection and per-stroke features can be exercised on synthetic signals, but the timing of acceleration peaks relative to a catch must be checked against an independent reference. The original cohort example averages a within-record standardized index, whose mean is zero by construction; that quantity cannot provide the claimed between-athlete ordering. An informative cohort comparison would require a fixed reference scale, explicit definitions, uncertainty and independent validation.

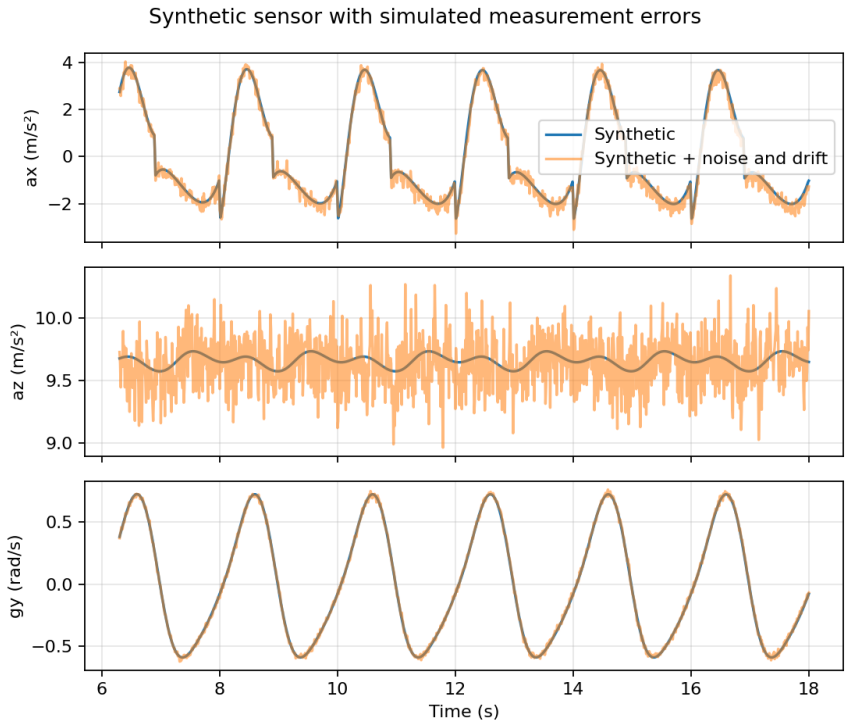


Figure 8: Simulated sensor channels with added synthetic measurement errors.

6 Reproducibility and validation plan

The accompanying scripts separate the original single-mass baseline, reconstructed slide and multi-body surrogate, IMU processing, synthetic fitting, and figure generation. The test suite checks the baseline energy residual under timestep refinement, unforced momentum conservation in the moving-mass model, slide derivatives, stationary gravity handling, event detection on periodic synthetic pulses, and parameter recovery under selected synthetic conditions. These checks establish consistency only within their stated assumptions; they are not experiments or proof that the numerical code reproduces every archived figure.

A measured validation study should document boat class, sensor mounting and axes, timing synchronization, calibration, environmental conditions, and independent speed and force references. The fitting protocol should fix or independently estimate confounded parameters, report the observables and weighting used, and assess sensitivity and uncertainty across independent strokes and sessions. Evaluation on held-out data is necessary to distinguish a model that explains training observations from one that predicts new conditions. Until such measurements are available, the model is best understood as a transparent framework for developing and testing hypotheses.

7 Conclusion

The baseline simulation illustrates how prescribed thrust and drag shape the speed and energy balance of a rowing shell during startup. The moving-mass, pitch, and sensor models extend that framework, but their parameters and predictions require independent measurement. The numerical and synthetic checks support the implementation under the stated assumptions; they do

not demonstrate a real-world efficiency gain. Future work should obtain synchronized force, speed, and inertial data, test sensitivity and identifiability, and evaluate predictions on held-out sessions.

A Model parameters and definitions

Table 1 records the baseline values used in the included simulation. The drag coefficient has units of kg/m, equivalent to $\mathrm{N\,s^2/m^2}$. These inputs define an illustrative case; they were not fitted to the measurements of a named crew.

Table 1: Baseline inputs for the 20-second startup simulation.

Quantity	Value	Unit
Mass	766.5	kg
Drag coefficient	40	kg/m
Drive force	3000	N
Drive duration	0.9	s
Stroke period	2.0	s
Time step	0.01	s
Initial speed	0	m/s

For the moving-mass prototype, the values in the archived listing are 250 kg for the hull-fixed mass, 500 kg for the moving mass, 0.85 m slide travel, 1200 $\mathrm{kg\,m^2}$ pitch inertia, 50,000 $\mathrm{N\,m/rad}$ pitch stiffness, 2,500 $\mathrm{N\,m\,s/rad}$ pitch damping, and 18,000 N of pitch-moment gain per meter of slide displacement. These values are effective parameters used to explore the equations; no fit or independent uncertainty estimate is available. The baseline and extended cases therefore do not use identical mass and force histories and should not be compared as if they were a controlled experiment.

B Energy and moving-mass derivations

Starting with $m\dot{v} = F - kv^2$ and multiplying by v yields $d(mv^2/2)/dt = Fv - kv^3$ for nonnegative v. Integrating between two specified times gives $\Delta K = W - D$. At an exactly repeated state, net kinetic-energy change over a whole cycle is zero even though both thrust work and drag work can be substantial. Consequently, the finite startup fraction $\Delta K/W$ changes with the observation window and cannot summarize rowing efficiency in steady operation.

For the extended translation model, the absolute crew position is $X_b + x_r$. Total translational momentum is $M\dot{X}_b + m_r\dot{x}_r$. Differentiation and the assumed external force balance yield the equation in the main text. In the absence of external forces the momentum remains constant, a property checked in the numerical tests. This relation does not by itself account for rotational energy, work by internal actuators, or the fluid forces on the oar.

C Scope of the synthetic sensor and fitting examples

The reconstructed synthetic sensor uses a planar approximation and a chosen gravity convention. The signal is generated from model states, and the corrupted version adds modeled noise, bias, and drift. This setup can check whether an analysis pipeline behaves consistently with its own assumptions, but it cannot estimate sensor accuracy, environmental variability, or physiological

effects. In particular, the small-angle filter assumes its acceleration input provides information about gravity, an assumption that may fail during strong linear acceleration.

The fitting example estimates two surrogate coefficients while holding other inputs fixed. A noiseless fit to data from the same model is an implementation check with a favorable model match. The drag-mismatch example illustrates that a successful optimizer termination does not ensure a correct physical estimate. Further work should examine different initial guesses, measurement noise levels, time windows, parameter correlations, and independent datasets before interpreting fitted coefficients. The original source versions, including unverified figures and listings, remain archived with the project for reference.

References

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[4] A. Mola, *Multi-physics and Multilevel Fidelity Modeling and Analysis of Olympic Rowing Boat Dynamics*, PhD thesis, Virginia Tech, 2010. `https://vtechworks.lib.vt.edu/server/api/core/bitstreams/4f7623b0-5169-4a45-a6e3-8b181601a912/content`.

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